GATE ME 90 Days Challenge

Day 39/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
Engineering Mathematics	**Differential Equations**	**66 days to GATE**

How to use this set: 15 questions (8 MCQ, 7 NAT) on first-order ODEs (separable, linear, exact, Bernoulli), higher-order linear ODEs with constant coefficients, complementary function and particular integral, the Euler–Cauchy equation, Laplace transforms and their use in initial-value problems, and the heat, wave and Laplace equations. Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.

Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 MCQ

The differential equation whose general solution is $y = A\cos 3x + B\sin 3x$ (A, B arbitrary constants) is

(A) $y'' - 9y = 0$ **(B)** $y'' + 3y = 0$

(C) $y'' + 9y = 0$ **(D)** $y' + 3y = 0$

Q2 NAT

If $dy/dx = xy$ and $y(0) = 1$, then $y(2)$ (round off to two decimal places) is _____

Q3 NAT

The solution of $x\,dy/dx + 2y = x^2$ $(x > 0)$ with $y(1) = 1$ is evaluated at $x = 2$. The value of $y(2)$ (round off to four decimal places) is _____

Q4 MCQ

The general solution of $y'' + 6y' + 9y = 0$ is

(A) $y = Ae^{-3x} + Be^{3x}$ **(B)** $y = (A + Bx)e^{-3x}$

(C) $y = A\cos 3x + B\sin 3x$ **(D)** $y = Ae^{-3x}$

Q5 NAT

The solution of $y'' - y = 0$ with $y(0) = 0$ and $y'(0) = 2$, evaluated at $x = 1$ (round off to three decimal places), is _____

Q6 MCQ

A particular integral of $y'' + 4y = \sin 2x$ is

(A) $-(x/4)\cos 2x$ **(B)** $(1/4)\sin 2x$

(C) $-(1/4)\cos 2x$ **(D)** $(x/4)\sin 2x$

Q7 NAT

For the Euler–Cauchy equation $x^2y'' + xy' - 4y = 0$ with $y(1) = 2$ and $y'(1) = 0$, the value of $y(2)$ is _____

Q8 MCQ

The Laplace transform of t sin t is

(A) $1/(s^2 + 1)^2$
(B) $2/(s^2 + 1)^2$
(C) $(s^2 - 1)/(s^2 + 1)^2$
(D) $2s/(s^2 + 1)^2$

Q9 NAT

Using Laplace transforms, solve $y' + 3y = e^{-t}$ with $y(0) = 0$. The value of y at t = 1 (round off to three decimal places) is _____

Q10 MCQ

If $F(s) = (3s + 1)/(s^2 + 2s + 5)$ is the Laplace transform of f(t), then $f(0^+)$ is

(A) 0
(B) 1/5
(C) 3
(D) ∞

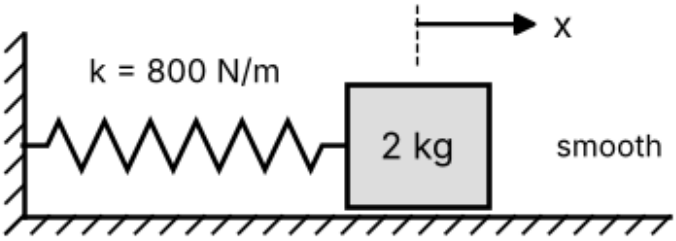
Q11 MCQ

The partial differential equation $u_{xx} + 2u_{xy} + 5u_{yy} = 0$ is

(A) parabolic
(B) elliptic
(C) hyperbolic
(D) none of these, since it has a mixed term

Q12 NAT

The 2 kg block shown is attached to a spring of stiffness 800 N/m on a smooth floor. It is pulled 10 mm to the right and released from rest, so that $2\ddot{x} + 800x = 0$, $x(0) = 0.01$ m, $\dot{x}(0) = 0$. The time at which the block first passes through the equilibrium position (in s, round off to four decimal places) is _____



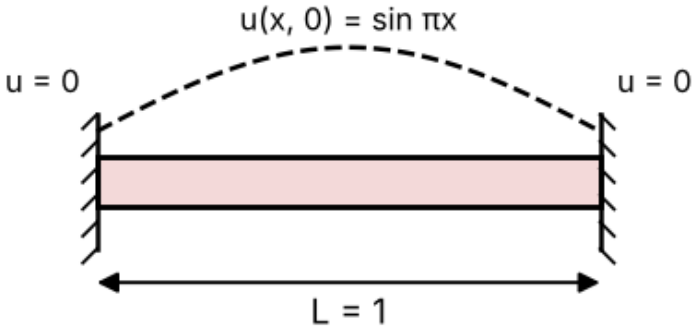
Q13 MCQ

The general solution of $(y \cos x + 2x\, e^y)\, dx + (\sin x + x^2e^y + 2)\, dy = 0$ is

(A) $y \sin x + x^2e^y + 2y = C$
(B) $y \sin x + x^2e^y = C$
(C) $y \cos x + x^2e^y + 2y = C$
(D) $\sin x + x^2e^y + 2y = C$

Q14 NAT

A rod of unit length, shown in the figure, obeys $u_t = u_{xx}$ with $u(0, t) = u(1, t) = 0$ and $u(x, 0) = \sin \pi x$. The temperature at the mid-point $x = 0.5$ at $t = 0.1$ (round off to three decimal places) is _____



Q15 MCQ

The Bernoulli equation $dy/dx + y = y^2$ is reduced to a linear equation by the substitution $v = 1/y$. The resulting equation is

(A) $dv/dx + v = 1$

(B) $dv/dx - v = -1$

(C) $dv/dx + v = -1$

(D) $dv/dx - v = 1$

Answer Key

Q	Type	Answer
Q1	MCQ	(C) $y'' + 9y = 0$
Q2	NAT	7.39 (accept 7.35 to 7.43)
Q3	NAT	1.1875
Q4	MCQ	(B) $y = (A + Bx)e^{-3x}$
Q5	NAT	2.350 (accept 2.338 to 2.362)
Q6	MCQ	(A) $-(x/4)\cos 2x$
Q7	NAT	4.25
Q8	MCQ	(D) $2s/(s^2 + 1)^2$
Q9	NAT	0.159 (accept 0.158 to 0.160)
Q10	MCQ	(C) 3
Q11	MCQ	(B) elliptic
Q12	NAT	0.0785 (accept 0.0781 to 0.0790)
Q13	MCQ	(A) $y \sin x + x^2e^y + 2y = C$
Q14	NAT	0.373 (accept 0.371 to 0.375)
Q15	MCQ	(B) $dv/dx - v = -1$

Detailed Solutions

Q1 **Answer: (C) $y'' + 9y = 0$**

Concept: Two arbitrary constants → eliminate them by differentiating twice; the result is a 2nd-order ODE.

$y' = -3A \sin 3x + 3B \cos 3x$, $y'' = -9A \cos 3x - 9B \sin 3x = -9y$

So **$y'' + 9y = 0$**.

Exam tip: Number of arbitrary constants = order of the ODE of the family.

Q2 **Answer: 7.39 (accept 7.35 to 7.43)**

Concept: Variables separable: $dy/y = x\,dx$.

$\ln y = x^2/2 + C$; $y(0) = 1$ gives $C = 0$, so $y = e^{x^2/2}$

$y(2) = e^2 =$ **7.39**

Exam tip: Always apply the initial condition before substituting the final x.

Q3 **Answer: 1.1875**

Concept: Write in standard linear form $dy/dx + P(x)y = Q(x)$ before finding the integrating factor.

$dy/dx + (2/x)y = x$, so IF $= e^{\int 2/x\,dx} = x^2$

$d(x^2y)/dx = x^3 \rightarrow x^2y = x^4/4 + C$; $y(1) = 1$ gives $C = 3/4$

$y = x^2/4 + 3/(4x^2)$, so $y(2) = 1 + 3/16 =$ **1.1875**

Exam tip: Divide by the coefficient of dy/dx first; using P = 2 instead of 2/x is the classic slip.

Q4 **Answer: (B) $y = (A + Bx)e^{-3x}$**

Concept: Auxiliary equation $m^2 + 6m + 9 = 0 \rightarrow (m + 3)^2 = 0$, a repeated root $m = -3$.

For a repeated root the second solution is $x\,e^{mx}$: **$y = (A + Bx)e^{-3x}$**.

In a spring–mass–damper this is the critically damped case ($\zeta = 1$).

Exam tip: Repeated root → multiply by x; complex roots $\alpha \pm i\beta \rightarrow e^{\alpha x}(A \cos \beta x + B \sin \beta x)$.

Q5 **Answer: 2.350 (accept 2.338 to 2.362)**

Concept: $m^2 - 1 = 0 \rightarrow m = \pm 1$, so $y = Ae^x + Be^{-x}$.

$y(0) = A + B = 0$ and $y'(0) = A - B = 2 \rightarrow A = 1, B = -1$

$y = e^x - e^{-x} = 2 \sinh x$

$y(1) = e - 1/e =$ **2.350**

Exam tip: With $y(0) = 0$, the hyperbolic form (sinh) is quicker than solving for A and B.

Q6 **Answer: (A) $-(x/4) \cos 2x$**

Concept: sin 2x is already a solution of the homogeneous equation (roots $\pm 2i$) — this is resonance, so multiply the trial by x.

Using $1/(D^2 + a^2) \sin ax = -(x/2a) \cos ax$ with $a = 2$:

$y_p =$ **$-(x/4) \cos 2x$** (check: substitution gives sin 2x ✓)

Exam tip: Resonant forcing makes the amplitude grow linearly with time — the reason operating near natural frequency is dangerous.

Q7 **Answer: 4.25**

Concept: Substitute $y = x^m$: $m(m - 1) + m - 4 = 0 \rightarrow m^2 = 4 \rightarrow m = \pm 2$.

$y = Ax^2 + Bx^{-2}$; $y(1) = A + B = 2$, $y'(1) = 2A - 2B = 0 \rightarrow A = B = 1$

$y(2) = 4 + 1/4 =$ **4.25**

Exam tip: For Euler–Cauchy the auxiliary equation is $m(m - 1) + am + b = 0$, not $m^2 + am + b = 0$.

Q8 **Answer: (D) $2s/(s^2 + 1)^2$**

Concept: Multiplication by t: $L\{t\,f(t)\} = -dF/ds$.

$L\{\sin t\} = 1/(s^2 + 1)$; $-d/ds\,[1/(s^2 + 1)] =$ **$2s/(s^2 + 1)^2$**

Exam tip: Each extra factor of t means one more derivative with respect to s, with a sign of (-1).

Q9 **Answer: 0.159 (accept 0.158 to 0.160)**

Concept: $L\{y'\} = sY - y(0)$; transform, solve for Y(s), then use partial fractions.

$sY + 3Y = 1/(s + 1) \rightarrow Y = 1/[(s + 1)(s + 3)] = ½[1/(s + 1) - 1/(s + 3)]$

$y(t) = ½(e^{-t} - e^{-3t})$

$y(1) = ½(0.36788 - 0.04979) =$ **0.159**

Exam tip: Check $y(0) = 0$ from your answer before moving on.

Q10 **Answer: (C) 3**

Concept: Initial value theorem: $f(0^+) = \lim_{s\to\infty} sF(s)$.

$sF(s) = (3s^2 + s)/(s^2 + 2s + 5) \to 3$ as $s \to \infty$

$f(0^+) = \mathbf{3}$. (Check: $f = e^{-t}(3\cos 2t - \sin 2t)$, $f(0) = 3$ ✓)

Exam tip: Initial value: $s \to \infty$. Final value: $s \to 0$, and only if all poles of $sF(s)$ are in the left half-plane.

Q11 **Answer: (B) elliptic**

Concept: For $Au_{xx} + Bu_{xy} + Cu_{yy}$: $B^2 - 4AC < 0$ elliptic, $= 0$ parabolic, > 0 hyperbolic.

$B^2 - 4AC = 4 - 20 = -16 < 0 \to$ **elliptic**

Exam tip: Laplace equation: elliptic; heat equation: parabolic; wave equation: hyperbolic. Here B is the full coefficient of u_{xy} (not B/2).

Q12 **Answer: 0.0785 (accept 0.0781 to 0.0790)**

Concept: $\ddot{x} + \omega^2 x = 0$ with $\omega = \sqrt{(k/m)}$.

$\omega = \sqrt{(800/2)} = 20$ rad/s; with $\dot{x}(0) = 0$, $x = 0.01\cos 20t$

First zero when $20t = \pi/2 \to t = \pi/40 =$ **0.0785 s** (a quarter period)

Exam tip: Released from rest at an extreme, the block reaches equilibrium after T/4, whatever the amplitude.

Q13 **Answer: (A) $y \sin x + x^2e^y + 2y = C$**

Concept: Check exactness: $\partial M/\partial y = \cos x + 2x\, e^y = \partial N/\partial x$ ✓

Integrate M with respect to x: $y \sin x + x^2e^y$; add the terms of N free of x: $2y$

Solution: $\mathbf{y \sin x + x^2e^y + 2y = C}$

Exam tip: After integrating M, add only those terms of N that do not contain x.

Q14 **Answer: 0.373 (accept 0.371 to 0.375)**

Concept: Separation of variables gives modes $\sin(n\pi x)\, e^{-n^2\pi^2 t}$; the initial profile contains only $n = 1$.

$u(x, t) = \sin(\pi x)\, e^{-\pi^2 t}$

$u(0.5, 0.1) = 1 \times e^{-0.98696} =$ **0.373**

Exam tip: Higher modes decay as n^2, so after a short time only the first mode matters.

Q15 **Answer: (B) dv/dx − v = −1**

Concept: For $dy/dx + Py = Qy^n$, put $v = y^{1-n}$; here $n = 2$, so $v = 1/y$.

$dv/dx = -y^{-2}\, dy/dx$. Divide the ODE by y^2: $y^{-2}dy/dx + y^{-1} = 1 \to -dv/dx + v = 1$

So $\mathbf{dv/dx - v = -1}$ (solution $v = 1 + Ce^x$, i.e. $y = 1/(1 + Ce^x)$).

Exam tip: Watch the sign that comes from differentiating 1/y.

What next?

Day 39 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

Website test series

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